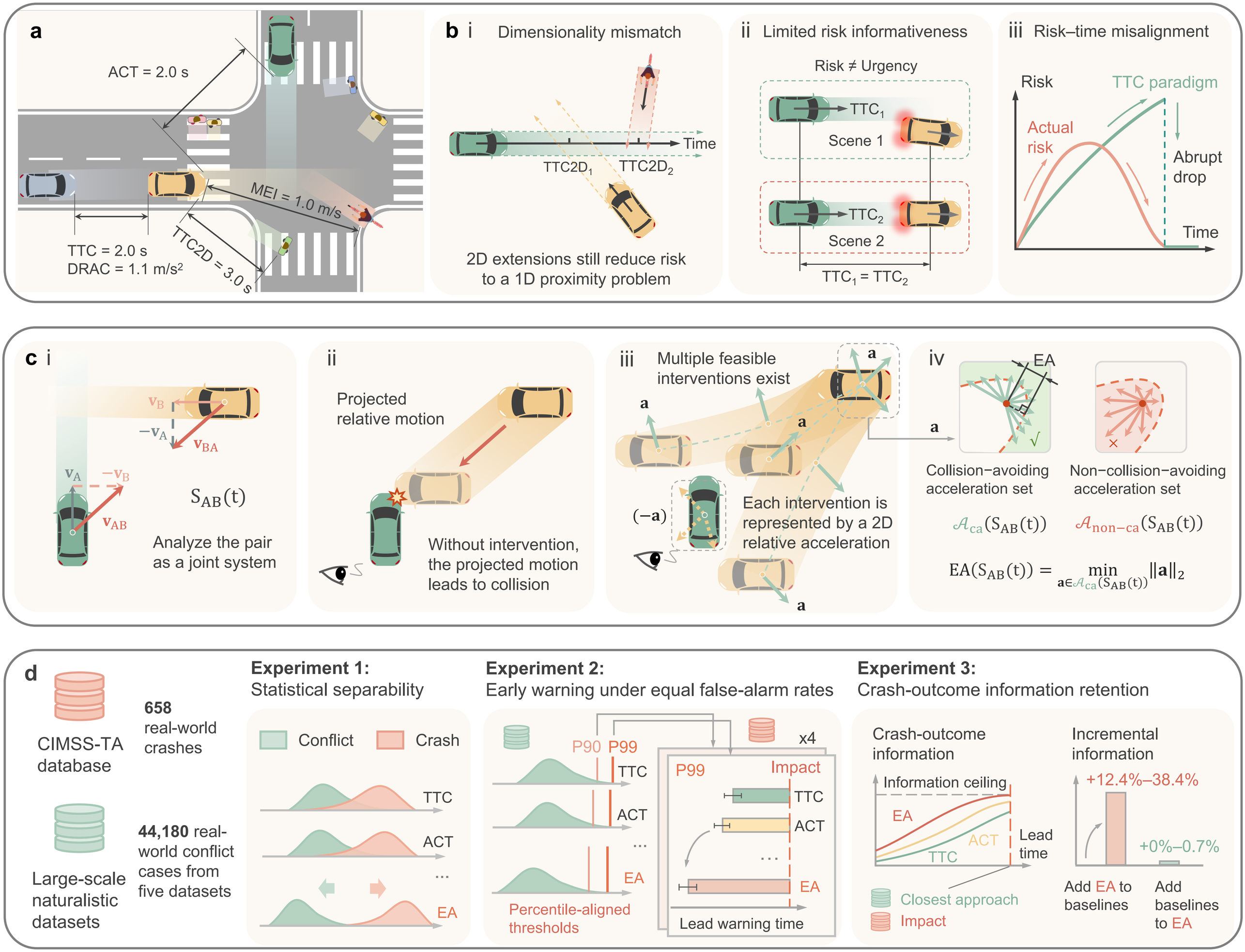
a
ACT = 2.0 s
MEI = 1.0 m/s
TTC = 2.0 s
DRAC = 1.1 m/s²
TTC2D = 3.0 s
b i Dimensionality mismatch
Time
TTC2D1
TTC2D2
2D extensions still reduce risk to a 1D proximity problem
ii Limited risk informativeness
Risk ≠ Urgency
TTC1
Scene 1
TTC2
Scene 2
TTC1 = TTC2
iii Risk–time misalignment
Risk
TTC paradigm
Actual risk
Abrupt drop
Time
c i
vB
−vA
vBA
vA
−vB
vAB
SAB(t)
Analyze the pair as a joint system
ii
Projected relative motion
Without intervention, the projected motion leads to collision
iii Multiple feasible interventions exist
a
(−a)
Each intervention is represented by a 2D relative acceleration
iv
EA
Collision−avoiding acceleration set
Non−collision−avoiding acceleration set
$\mathcal{A}_{ca}(S_{AB}(t))$
$\mathcal{A}_{non-ca}(S_{AB}(t))$
$EA(S_{AB}(t)) = \min_{a \in \mathcal{A}_{ca}(S_{AB}(t))} \|a\|_2$
d
CIMSS-TA database
658 real-world crashes
Large-scale naturalistic datasets
44,180 real-world conflict cases from five datasets
Experiment 1: Statistical separability
Conflict
Crash
TTC
ACT
EA
Experiment 2: Early warning under equal false-alarm rates
P90
P99
x4
Impact
Percentile-aligned thresholds
Lead warning time
Experiment 3: Crash-outcome information retention
Crash-outcome information
Information ceiling
Lead time
Closest approach
Impact
Incremental information
+12.4%–38.4%
+0%–0.7%
Add EA to baselines
Add baselines to EA